Machine Learning Final Project Report (Phase 1)

Repository: https://github.com/omarelnainay/ML_Project

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1. Project Overview

This project demonstrates a complete end-to-end machine learning pipeline applied to both classification and regression problems.

• Classification: Predicting passenger survival on the Titanic

• Regression: Predicting diamond prices based on physical attributes

2. Datasets Used

2.1 Titanic Dataset

Source: Kaggle

Target: Survived (0 = No, 1 = Yes)

Features: Pclass, Sex, Age, SibSp, Fare, Embarked

2.2 Diamonds Dataset

Source: Kaggle

Target: price

Features: carat, cut, color, clarity, depth, table, x, y, z

3. Project Structure

- Titanic-Dataset.csv – Raw dataset
- diamonds.csv – Raw dataset
- titanic_visualization.py
- diamond_visualization.py
- Preprocess_titanic.py
- Preprocess_diamond.py
- DecisionTreeClassifier.py

- ClassificationAdaboost.py
- LinearRegression.py
- DecisionTreeRegressor.py
- RegressionAdaboost.py
- README.md

4. Methodology

4.1 Data Exploration

Checked structure, missing values, statistics, and correlations.

4.2 Data Preprocessing

Handled missing values, removed duplicates, encoded categorical variables, and scaled features.

4.3 Visualization

Used histograms, scatter plots, and box plots (Matplotlib).

4.4 Modeling

Classification: Decision Tree, AdaBoost

Regression: Linear Regression, Decision Tree, AdaBoost

4.5 Evaluation

Classification: Accuracy, Precision, Recall, F1-score

Regression: MAE, MSE, RMSE, R^2

5. Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

6. Key Findings

Titanic: preprocessing and encoding improved performance.

Diamonds: scaling and correlations were important.

7. Conclusion

The project demonstrates a full ML pipeline with strong emphasis on preprocessing and model comparison.